



Intelligent Decision Support System for Sri Lankan Tea Estate Management

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Problem Introduction — Video



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Problem Overview



Delayed disease detection

- Tea leaf diseases are often detected late because inspections are done manually.
- This delay allows pests and diseases to spread across the plantation.



Inefficient labor allocation

- Tea estate workers are often scheduled without proper data or planning.
- This can cause too many workers in some areas and not enough in others.



Inconsistent tea sorting & grading

- Tea grading is usually done manually by workers.
- Different judgments can lead to inconsistent tea quality



Lack of tea-specific weather intelligence

- Tea plantations rely on general weather forecasts that are not specific to their fields.
- This makes it difficult to plan irrigation, spraying, and harvesting.

Users & Stakeholders



Estate Managers

Use the dashboard to monitor crop health, manage workers, and plan plantation operations.

Supervisors

Receive alerts, task lists, and mobile guidance to manage field activities.

Estate Owners

Monitor plantation performance, profits, and market quality standards.

Factory Operators

Use quality forecasts and leaf supply predictions to plan processing operations.

Plantation Workers

Receive clear daily tasks, safety alerts, and guidance from the system.

Impact of the Problem

Economic Loss

Reduced tea yield and lower leaf quality decrease estate revenue and profits.

Operational Inefficiencies

Poor planning leads to wasted labor, incorrect chemical use, and higher production costs.

Social Consequences

Unstable income and heavy workloads affect plantation workers and local communities.



Existing Solutions & Limitations



Manual Inspections

High labor cost and limited coverage.
Disease detection may be slow or inconsistent.

Generic Weather Apps

Provide general forecasts but not plantation-specific weather insights.

Traditional Tea Grading

Manual grading depends on human judgment and may cause inconsistent results.

Isolated Technologies

Sensors or drones often work separately without an integrated system.

Proposed Solution

An AI-driven Decision Support System integrating machine learning, computer vision, and predictive analytics to deliver:



Image-Based Disease Detection

Use camera images and computer vision models to detect tea leaf diseases early.



Predictive Analytics

Analyze historical, sensor, and weather data to predict yield and tea quality.



Optimized Labor Scheduling

Use data to allocate workers efficiently and reduce idle time.



Microclimate Intelligence

Provide local weather insights to support irrigation, spraying, and harvesting decisions.

Solution Overview — System Architecture

Intelligent Decision Support System for Sri Lankan Tea Estate Management



System Components



Tea Leaf Disease Detection

Detect common tea leaf diseases using image analysis and machine learning models. Early identification helps reduce crop damage and supports timely treatment.



Labor Management Optimization

Optimize worker scheduling and task allocation based on plantation needs. This helps improve harvesting efficiency and reduce idle labor time.



AI-Based Tea Sorting and Grading

Automatically classify tea leaf grades using computer vision techniques. This ensures consistent quality standards and reduces manual grading effort.



Tea-Specific Weather Intelligence

Analyze local weather data to provide plantation-specific forecasts. Helps plan irrigation, spraying, and harvesting activities more effectively.

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Tea Leaf Disease Detection and Pattern Analysis

INFORMATION TECHNOLOGY SPECIALIZATION

Component 1: Tea Leaf Disease Detection and Pattern Analysis

Research Question

- Current tea disease monitoring in many estates depends on manual visual inspection by field supervisors.
- Early symptoms of diseases such as Blister Blight, Red Rust, and Grey Blight are difficult to detect in large plantations.
- Delayed detection results in crop loss, reduced tea quality, and increased pesticide usage.

Specific Objective

To design and develop an **AI-based tea leaf disease detection system** that uses **image processing and Convolutional Neural Networks (CNN)** to identify diseases, estimate severity levels, and provide treatment recommendations for tea estates.

Sub Objectives

- Collect and label tea leaf disease images from tea estates.
- Preprocess images and build a **CNN-based disease classification model**.
- Detect **disease type and severity level** using image analysis.
- Generate **treatment recommendations** based on detected diseases.
- Analyze historical disease data to identify **seasonal and location-based patterns**.
- Provide **real-time alerts and reports** for estate managers.

Tea Leaf Disease Detection and Pattern Analysis

Research Gap

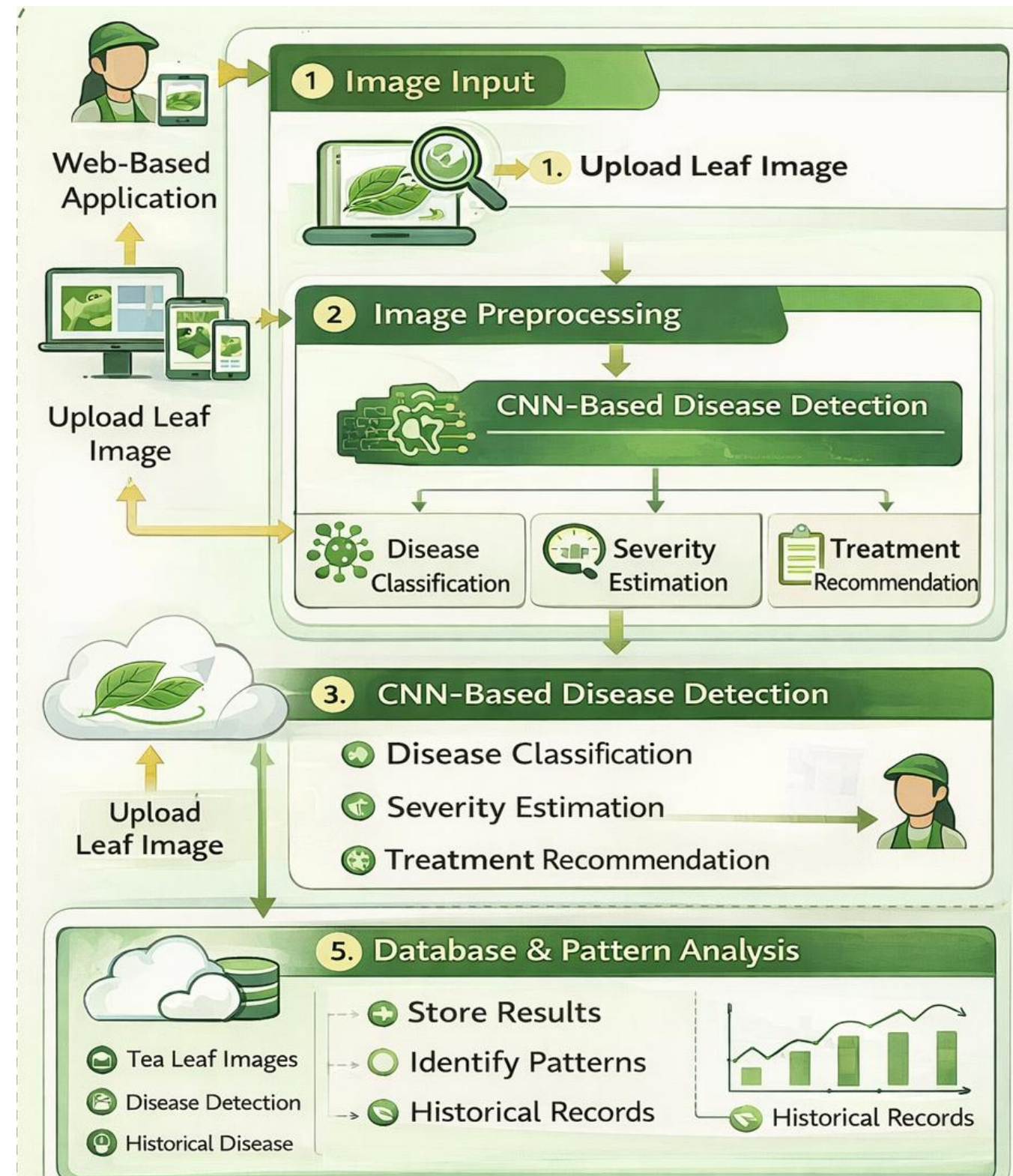
Feature	Existing Systems	Proposed Research
Automated Tea leaf Detection	X	✓
Tea-Specific Dataset	✓	✓
Disease Severity Estimation	X	✓
Treatment Recommendation System	X	✓
Disease Pattern Analysis	X	✓
Real-Time Estate Monitoring	X	✓

Tea Leaf Disease Detection and Pattern Analysis

- Tea leaf image data collection from estates and public datasets
- Image preprocessing and augmentation for model training
- CNN-based deep learning model development for disease classification
- Severity estimation and disease pattern analysis
- Web-based system integration for real-time monitoring
- Dashboard visualization for estate managers

Tea Leaf Disease Detection and Pattern Analysis

CNN-based tea leaf disease detection system with image input, severity analysis, and treatment recommendation.



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Labor Management & Welfare Optimization

INFORMATION TECHNOLOGY SPECIALIZATION

Component 2: Labor Management & Welfare Optimization

Research Question

- Labor management in many tea estates is handled using **manual attendance registers and supervision.**
- This makes it difficult to monitor **worker productivity, workload distribution, and welfare conditions.**
- Lack of data-driven monitoring can lead to **worker fatigue, inefficient task allocation, and reduced productivity.**

Specific Objective

To design and develop an **AI-based labor management system** that analyzes workforce data to improve **task allocation, productivity, and worker welfare monitoring**.

Sub Objectives

- Collect worker data including **attendance, workload, and health indicators**.
- Analyze labor performance using **machine learning techniques**.
- Predict **worker fatigue and workload imbalance**.
- Optimize **task allocation** based on worker availability and performance.
- Provide **alerts and insights** for estate managers through a dashboard.
- Improve **labor welfare monitoring and grievance management**.

Labor Management & Welfare Optimization

Research Gap

Feature	Existing Systems	Proposed Research
Digital Attendance Tracking	✓	✓
Workforce Productivity Analysis	✗	✓
AI-Based Task Allocation	✗	✓
Worker Fatigue Prediction	✗	✓
Welfare Monitoring	✗	✓
Integrated Decision Support	✗	✓

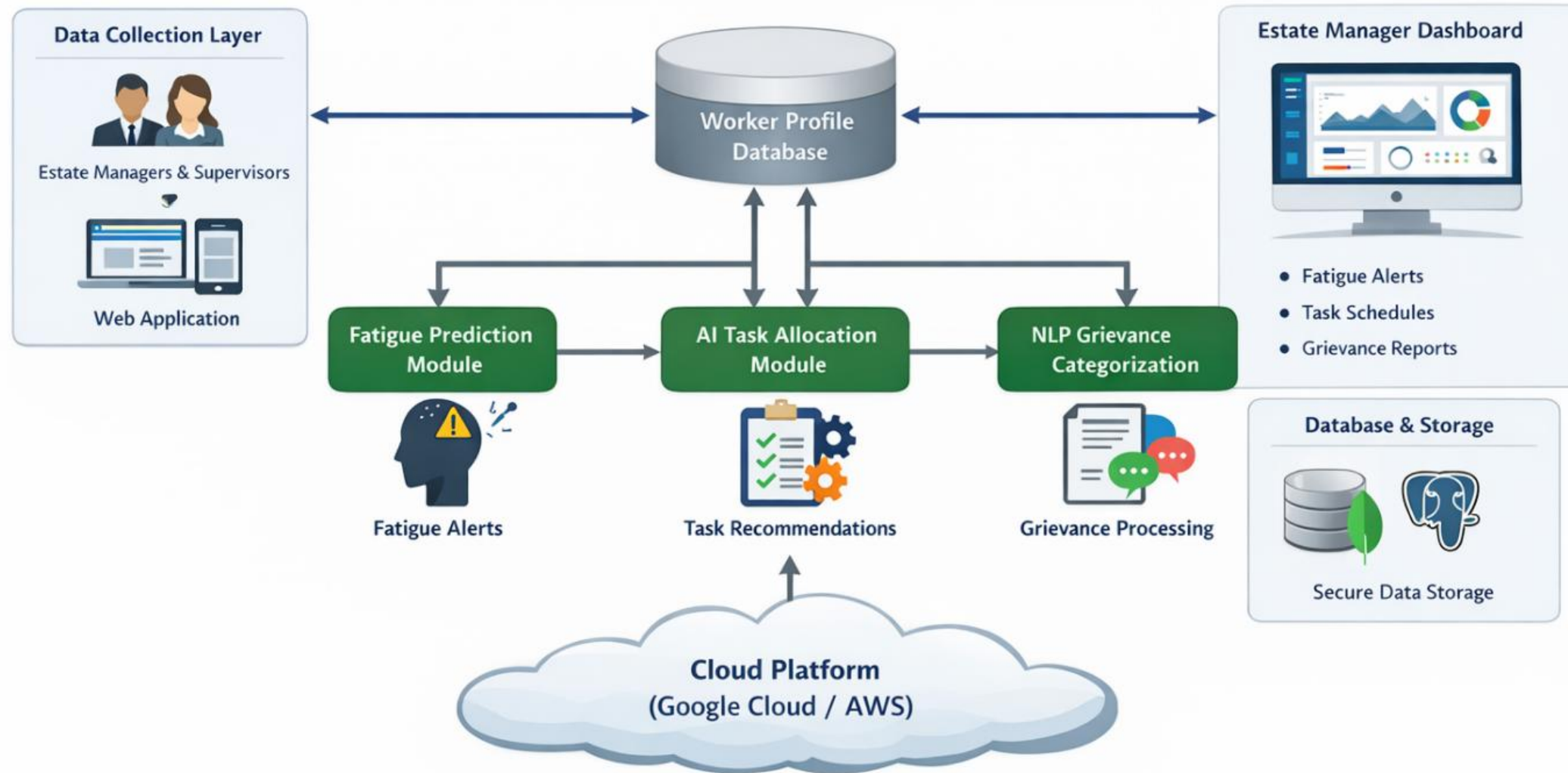
Labor Management & Welfare Optimization

- Collect worker data including attendance, health conditions, and task records.
- Preprocess and organize workforce data for analysis.
- Apply machine learning models for workforce analysis and fatigue prediction.
- Develop AI-based task allocation system.
- Integrate results into a web-based estate management dashboard.
- Provide real-time insights and recommendations for estate managers

Labor Management & Welfare Optimization

AI-Driven Decision Support System for Sri Lankan Tea Estate Management

Labor Management & Welfare Optimization System



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AI-Based Tea Sorting and Grading Automation

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AI-Based Tea Sorting and Grading Automation

Research Question

- Tea sorting and grading in many factories are performed manually by workers.
- Manual grading depends on human experience and visual judgment, which can lead to inconsistency.
- Large production volumes make accurate and consistent grading difficult.

AI-Based Tea Sorting and Grading Automation

Specific Objective

To design and develop an **AI-based tea sorting and grading system** that automatically classifies tea grades using **image processing and machine learning techniques**.

Sub Objectives

- Collect and label tea grade images from factory environments.
- Preprocess tea images to enhance **color, texture, and shape features**.
- Develop a **CNN-based model** for tea grade classification.
- Automatically classify grades such as **Dust, Pekoe, BOP, and FBOP**.
- Generate **batch-level grading reports** for factory management.
- Compare **manual grading performance** with **AI-based grading results**.

AI-Based Tea Sorting and Grading Automation

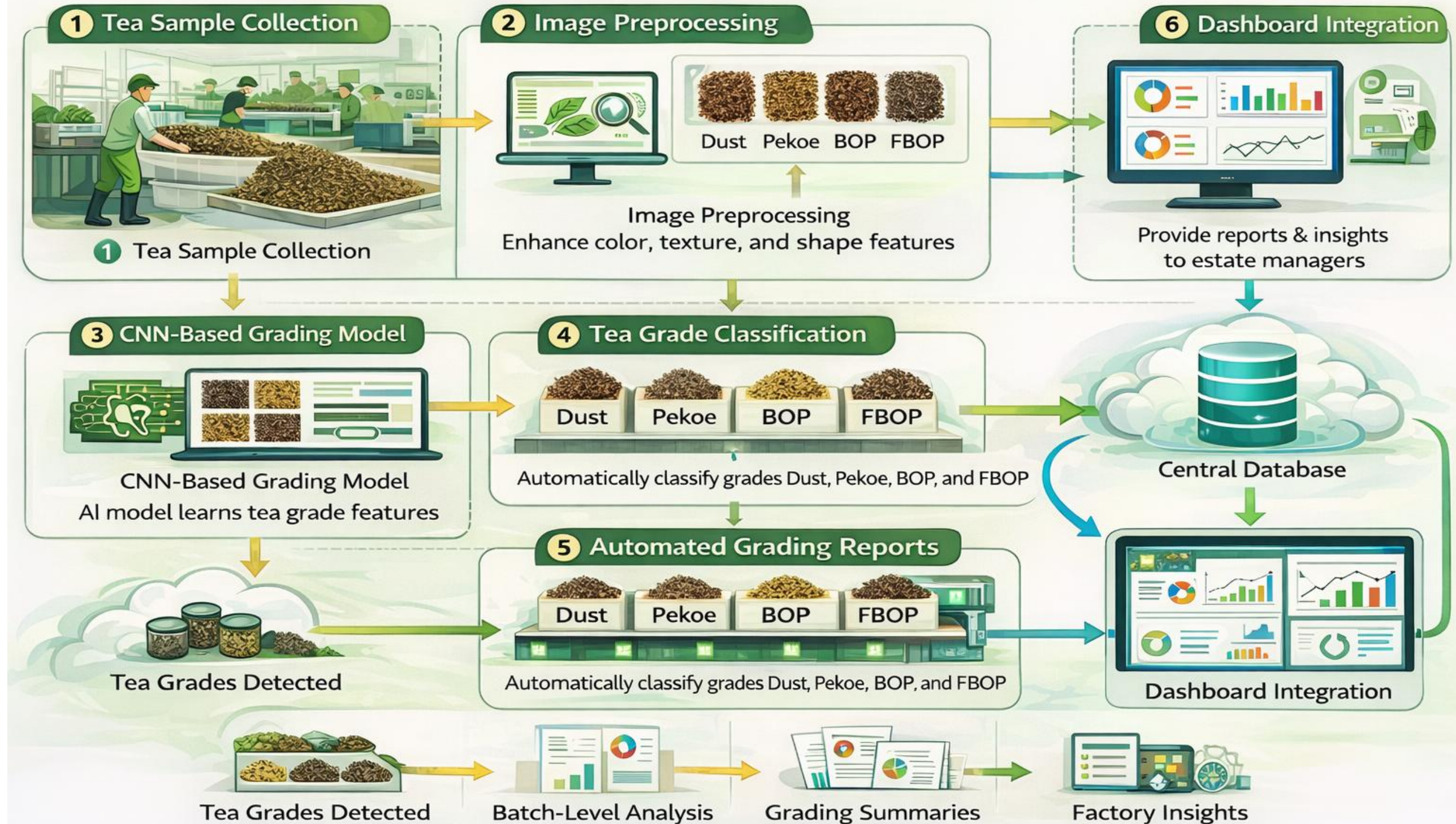
Research Gap

Feature	Existing Systems	Proposed Research
Manual Tea Sorting	✓	✓
Image-Based Tea Classification	✗	✓
AI-Based Tea Grade Prediction	✗	✓
Consistent Batch-Level Analysis	✗	✓
Automated Reporting	✗	✓
Integration with Estate Management Systems	✗	✓

AI-Based Tea Sorting and Grading Automation

- Collect tea sample images from factory sorting processes.
- Perform image preprocessing and feature extraction.
- Train a **CNN-based** deep learning model for tea grade classification.
- Evaluate classification accuracy for **Dust, Pekoe, BOP, and FBOP**.
- Generate **automated** grading reports and production summaries.
- Integrate results into the **estate management dashboard** for decision support.

AI-Based Tea Sorting and Grading Automation



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Tea-Specific Weather & Climate Intelligence System

INFORMATION TECHNOLOGY SPECIALIZATION

Tea-Specific Weather & Climate Intelligence System

Research Question

- Tea production is highly affected by weather conditions such as rainfall, humidity, and temperature.
- Estate managers currently rely on generic weather apps that provide only basic forecasts.
- These systems do not translate weather data into tea-specific cultivation decisions.

Tea-Specific Weather & Climate Intelligence System

Specific Objective

To design and develop a tea-specific weather and climate intelligence system that uses machine learning models to analyze weather patterns and provide proactive cultivation recommendations.

Sub Objectives

- Collect real-time weather data from Open Weather, Google Weather, and Visual Crossing APIs
- Process and normalize climate data for analysis
- Apply LSTM models for weather pattern prediction
- Use Random Forest to predict tea disease risks
- Generate irrigation recommendations using evapotranspiration calculations
- Provide alerts and cultivation insights through a web-based dashboard

Tea-Specific Weather & Climate Intelligence System

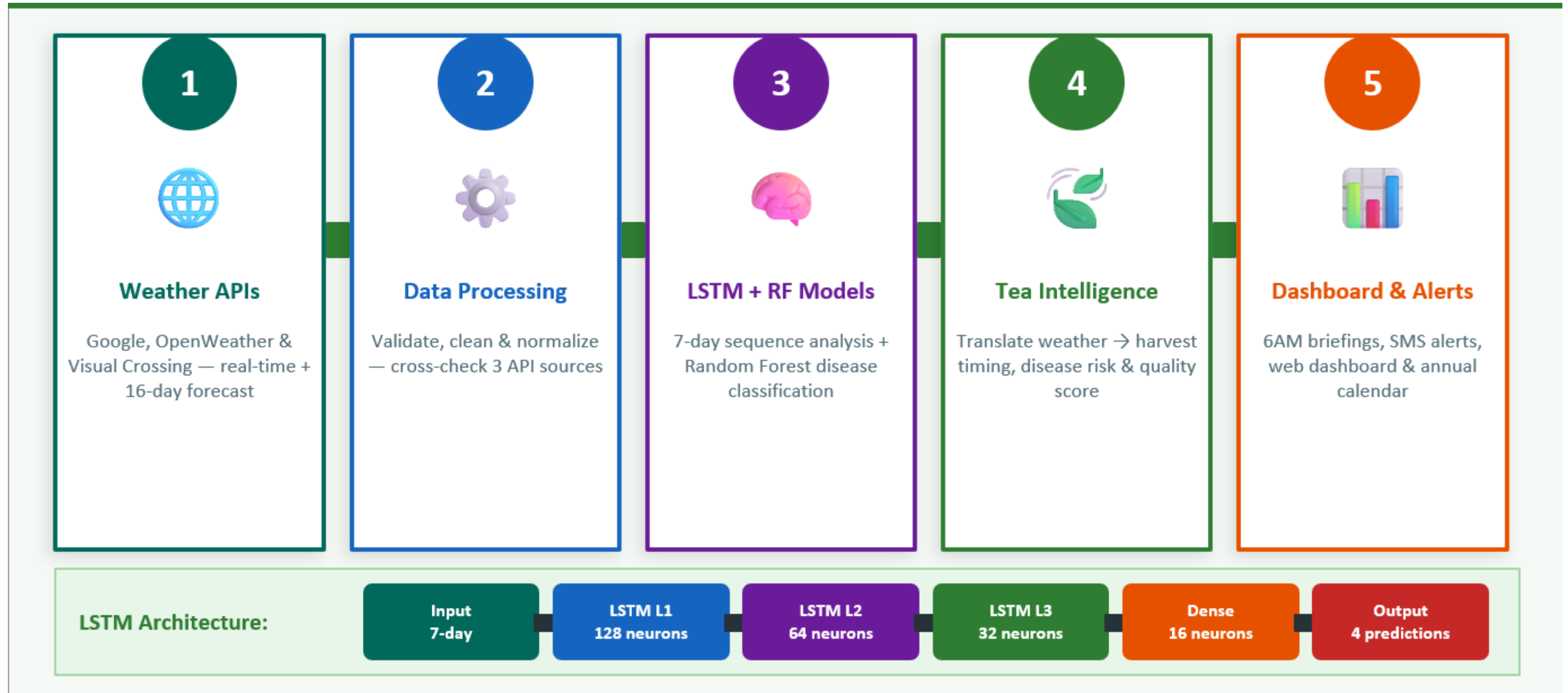
Research Gap

Feature	Existing Systems	Proposed Research
Basic Weather Forecast	✓	✓
Tea-Specific Climate Guidance	✗	✓
Disease Risk Prediction	✗	✓
AI-Based Yield Prediction	✗	✓
Irrigation Optimization	✗	✓
Plantation Decision Support	✗	✓

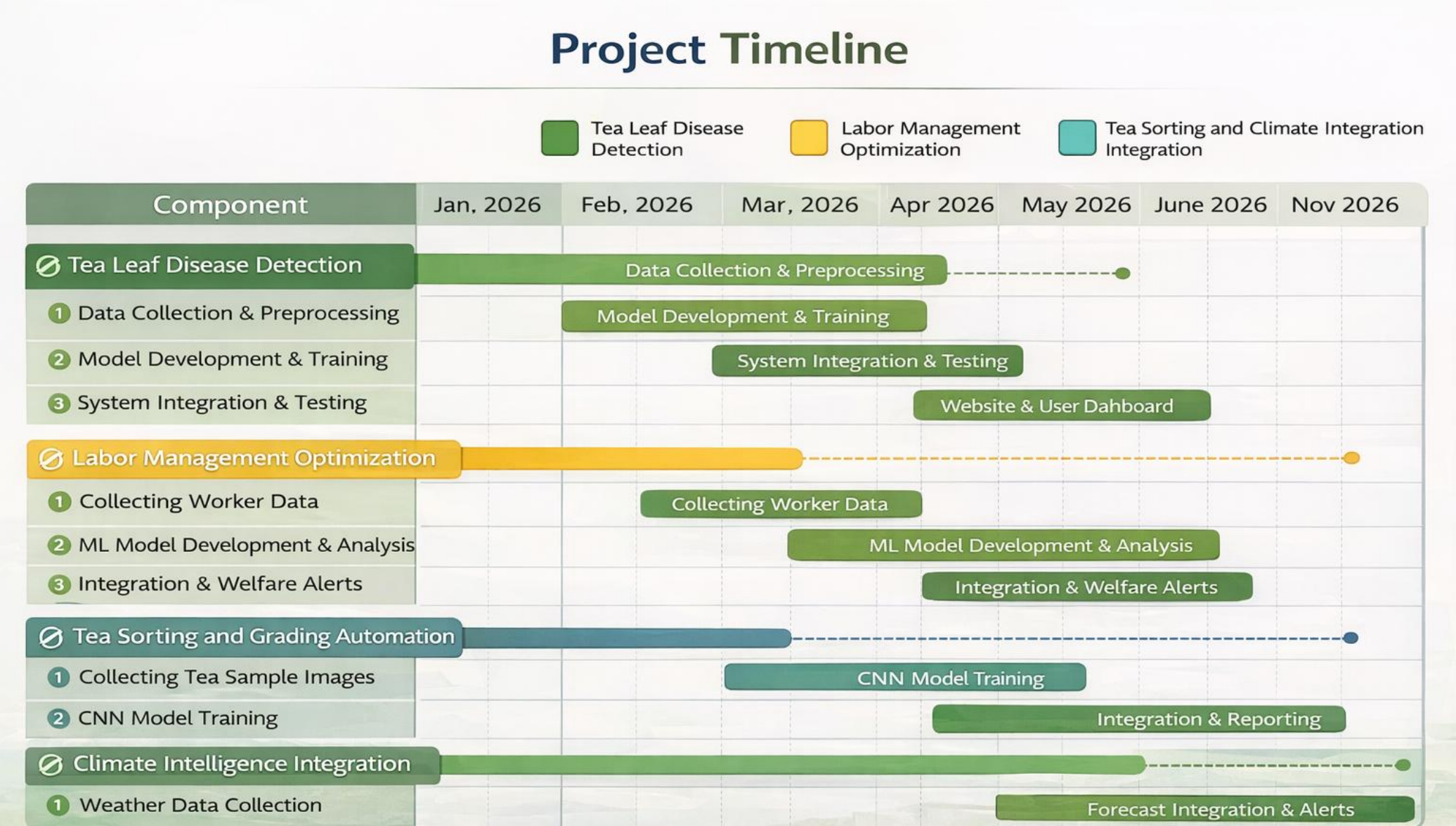
Tea-Specific Weather & Climate Intelligence System

- Collect weather data using multiple weather APIs
- Perform data cleaning and normalization
- Train LSTM models for weather pattern prediction
- Apply Random Forest models for disease risk analysis
- Generate irrigation and harvest recommendations
- Display insights through a web-based estate management dashboard

Tea-Specific Weather & Climate Intelligence System



Project Timeline (Mar → Nov)



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